ikerlan

MEMBER OF BASQUE RESEARCH & TECHNOLOGY ALLIANCE

SandGrain Cloud PoC – Deployment Manual (Cloud App MQTT + SDK).

Made by
Denis Stefanescu, Markel Baskaran
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OVERVIEW

This repository (sdk_cloud) contains a complete PoC environment for SandGrain mutual authentication, including:

- **EMQX MQTT Broker** (message transport layer)
- **Cloud PoC Application** (application_PoC_MQTT)
- **SandGrain Cloud Python SDK** (sandgrain_cloud_sdk)

The cloud application listens to device requests over MQTT and interacts with the SandGrain Device API to perform:

- requestCW
- replyRW
- requestRW (for mutual authentication)
- checkAuthStatusCloud

All components can be deployed together using Docker Compose.

REPOSITORY STRUCTURE

The repository is organized as follows:

- **application_PoC_MQTT/**
  Main runnable cloud service (MQTT listener + handlers)
- **sandgrain_cloud_sdk/**
  Internal Python SDK package used by the cloud app
- **.env (path: application_PoC_MQTT/.env)**
  Environment configuration (API endpoints, credentials, MQTT host, etc.)

  As an example: cp .env.example .env

  PLEASE MODIFY THIS FILE (.env) WITH THE CORRESPONDING CREDENTIALS
- **device_credentials.JSON (path: application_PoC_MQTT/device_credentials.JSON)**
  Environment configuration (API endpoints, credentials, MQTT host, etc.)

  As an example: cp device_credentials_example.json device_credentials.json

  PLEASE MODIFY THIS FILE (device_credentials.JSON) WITH THE CORRESPONDING CREDENTIALS
- **docker-compose.yml**
  Runs both EMQX broker and the cloud application
- **Dockerfile**
  Builds the cloud application container

DOCKER IMAGE BUILD (CLOUD APP)

The cloud application container is built **automatically** using the **Dockerfile**:

- Python base: python:3.12
- Installs required dependencies from the `requirements.txt`
- Installs the Cloud SDK
- Starts the MQTT cloud service

CONFIGURATION VIA .ENV

The system is configured through the provided .env file.

PLEASE MODIFY THIS FILE (.env) WITH THE CORRESPONDING CREDENTIALS

This file must include, at minimum:

- MQTT broker address and credentials
- SandGrain Device API base URL
- Cloudflare Access credentials (if enabled)
- Device login credentials JSON path

The Docker Compose setup automatically loads this .env.

RUNNING THE FULL SYSTEM (BROKER + CLOUD APP)

To start everything on one host:

Step 1 — Build and launch containers

```
docker compose up -d --build
```

This will start:

- emqx-enterprise (MQTT broker)
- sandgrain-cloud-app (PoC)

EXPECTED RUNTIME BEHAVIOR

Once running, the cloud application will:

- Subscribe to wildcard topic: `sandgrain/v1/+/req/+`
- Receive requests from embedded devices
- Dispatch actions through handlers (handlers.py)

- Call SandGrain Device API endpoints via the SDK
- Publish responses back to: `sandgrain/v1/<device_id>/rsp/<action>`